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High-temperature oxidation behaviour of a single-phase $(Mo,Ti)_5Si_3$ (Mo–Si–Ti) alloy

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The alloy composition Mo–37Si–40Ti (at.%) was synthesized by arc melting in order to obtain nearly single-phase $(Mo,Ti)_5Si_3$. At 750–1300 °C, the intermetallic compound was found to be superior to both single-phase Mo_5Si_3 and single-phase Ti_5Si_3 with regard to oxidation resistance. Upon oxidation, formation of a protective SiO_2–TiO_2 duplex scale was observed, which dramatically reduces outward diffusion of Ti. The results indicate that Ti alloying is a promising concept to further improve oxidation resistance of Mo-based alloys such as Mo–Si–B.
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Systematic examinations of refractory metal-rich silicide phases have revealed their outstanding structural features [1]. Recently, it was found by thermodynamic modeling with coupled experiments that (macro)alloying of Ti to the Mo–Si–B system stabilizes/extends the three-phase equilibrium Mo_ss–Mo_3Si–Mo_5SiB_2, where Mo_ss is the body-centered cubic Mo solid-solution phase. Moreover, if Ti is added in sufficiently high concentration, instead of the Mo_3Si phase, Mo_5Si_3 with Ti in solid solution is stabilized (in the following denoted as $(Mo,Ti)_5Si_3$) resulting in a Mo_ss–$(Mo,Ti)_5Si_3$–Mo_5SiB_2 alloy [2]. Here, Ti substitutes Mo, while the concentrations of Si or B in the existing phases are either unaffected or remain negligibly small [3]. Ti alloying may be important in the further design of three phase Mo–Si–B alloys for ultrahigh-temperature applications due to the following reasons: (i) (partially) substituting Mo by Ti will lead to a significant reduction in the alloy density (which is 9.6 g cm^{-3} for a typical composition of Mo–9 at.% Si–8 at.% B [4]), yielding a strong increase of turbine efficiency; (ii) in terms of oxidation resistance, replacing Mo_3Si by the $(Mo,Ti)_5Si_3$ in Mo–Si–B alloys may improve the overall high-temperature oxidation resistance of the alloy due to a significantly higher Si concentration [5].

It is well known that both Mo_3Si and Mo_5Si_3 in single-phase condition are only oxidation resistant at ultrahigh temperatures beyond 1600 °C due to a highly porous silica scale which forms as a consequence of the evaporation of MoO_3 [6]. Ti as a major alloying element might have a positive effect. The oxidation behaviour of single-phase Ti_5Si_3 as reference material has been previously investigated by Williams and Akinc [7] as well as by Mitra and Rao [8], and provides a basis for understanding the fundamental oxidation behaviour of $(Mo,Ti)_5Si_3$ too.

To the best of our knowledge, no data on the oxidation behaviour of the Mo–Si–Ti system has been reported. Hence, in this paper we investigate the oxidation behaviour of the alloy Mo–37Si–40Ti in the nearly single-phase $(Mo,Ti)_5Si_3$ condition in the temperature range between 750 and 1300 °C.

Guided by thermodynamic modeling [2], a molybdenum silicide alloy with the composition Mo–37Si–40Ti (at.%) was prepared from elemental powder mixtures of Mo, Si and Ti of 99.95%, 99.9% and 99.5% purity by an arc-melting process route. After drop casting the alloy, annealing under H_2 atmosphere was carried out at 1550 °C for 100 h followed by a subsequent

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homogenization treatment in Ar atmosphere at 1600 °C for 100 h, yielding the microstructure given in Figure 1. The alloy is (nearly) single-phase and comprises a Mo-rich $(Mo,Ti)_5Si_3$ phase with traces of Ti-rich $Ti(Mo)_5Si_3$, possibly due to fact that thermodynamic equilibrium was not achieved completely during the homogenization treatment.

The oxidation kinetics was studied under isothermal testing conditions by using continuous thermogravimetric analysis (TGA) in static laboratory air. The tailor-made system comprises a Rubotherm high-resolution magnetic suspension balance (10^{-5} g) equipped with automatic electronic drift compensation. Surface and cross-sectional scanning electron microscopy (SEM) in secondary-electron (SE) and backscattered-electron (BSE) mode was applied using a Helios Nanolab 600 FE-SEM (FEI) as well as energy-dispersive X-ray spectroscopy (EDS) to qualitatively determine the elements present in the scale with an Apollo XL silicon drift detector (EDAX) ensuring high count rates even for lightweight elements, e.g. O. Oxidation samples having the dimensions $10 \times 4 \times 3$ mm^3 were machined using a slow-cutting diamond saw. Prior to oxidation, the samples were ground and polished with SiC paper down to 1200 grit and ultrasonically cleaned in ethanol. For kinetics examination, TGA measurements were carried out up to 100 h with the samples hanging on an alumina hook connected contactlessly to the microbalance via a permanent magnet at the reaction chamber side as well as an electromagnet at the microbalance side. Oxidation experiments were performed in the temperature range 750–1300 °C.

The specific weight change vs. time for the alloy $(Mo,Ti)_5Si_3$ at 750, 820, 1100 and 1300 °C is shown in Figure 2. At all temperatures, continuous weight gaining kinetics with the oxidation rate increasing disproportionally with temperature is observed. At 1100 and 1300 °C, a strong linear oxidation relationship dominates the oxidation kinetics within the first hour of oxidation. This is in good agreement with the oxidation behaviour of elemental Ti at temperatures above 900 °C, which was reported by Kubaschewski [9].

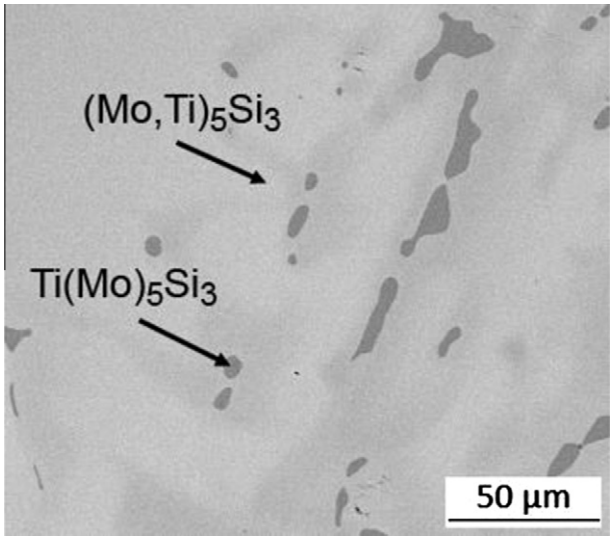


Figure 1. SEM microstructure of the arc melted alloy Mo–37Si–40Ti (SEM, BSE mode).

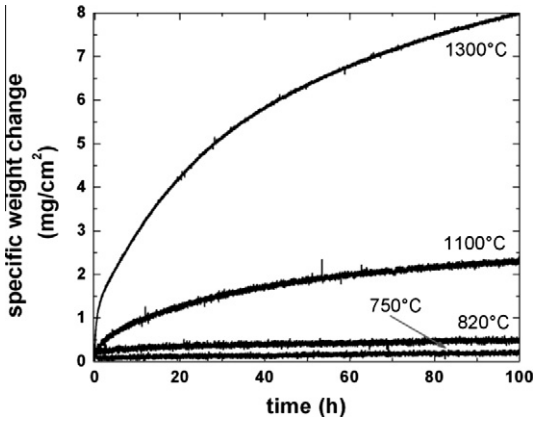


Figure 2. Specific weight change vs. time for the alloy Mo–37Si–40Ti oxidized in air at temperatures between 750 and 1300 °C.

Figure 3 shows the SEM microstructures (surface and cross-section views) of oxidized near single-phase $(Mo,Ti)_5Si_3$ after oxidation at 750 °C (surface, top left), 1100 °C (cross-section, top middle) and 1300 °C (surface, top right) for 100 h including the main features of oxide scale evolution. At 750 °C but also at 820 °C (not shown in the figure), nanometer-sized TiO_2 particles are found randomly distributed at the oxide/air interface without visible formation of a compact titania scale. In between and below the particle gaps, SiO_2 is detected. Continuous TiO_2 scale formation is observed at 1100 and 1300 °C. The structure of the TiO_2 grains exhibits a rather sharply edged structure. In this context, the relatively wide spacing of the TiO_2 grain boundaries at very high temperatures (see Fig. 3 (top right)) needs to be mentioned. It is assumed that the oxygen readily permeates through this open structure.

Based on the oxidized alloy cross-section after oxidation exposure at 1100 °C for 100 h given in Figure 3 (top middle), the oxide morphology can be subdivided into three major regions. (i) The outermost layer at the oxide/air interface consists of a single-phase and continuous TiO_2 scale exhibiting an average thickness of 4–8 μm at 1100 °C after 100 h. (ii) Below the titania layer, a duplex layer containing SiO_2 as well as TiO_2 is observed. Since TiO_2 exists as discontinuous and embedded particles within this scale, continuous SiO_2 is considered as the matrix or skeleton phase of this layer. (iii) At the duplex-layer/substrate interface, internal oxidation of Si is observed, but the presence of Ti was not confirmed. The SiO_2 particles are embedded within a Mo-rich matrix. EDS analyses underline the depletion of Ti within the internal oxidation zone. At 1300 °C, the oxide morphology cross-section is rather similar to the one found for 1100 °C, although the TiO_2 layer exhibits an average thickness of 80 μm. At the lower temperatures of 750 and 820 °C, the oxide layers are comparably thinner, exhibiting an identical morphological structure. The external TiO_2 scale possesses an average thickness of 200 nm, whereas the thickness of the duplex layer lies in the 500 nm range. The internal oxidation depth after 100 h is in the range of 1–2 μm.

In the following, the main factors determining oxidation kinetics are discussed briefly. The mechanisms

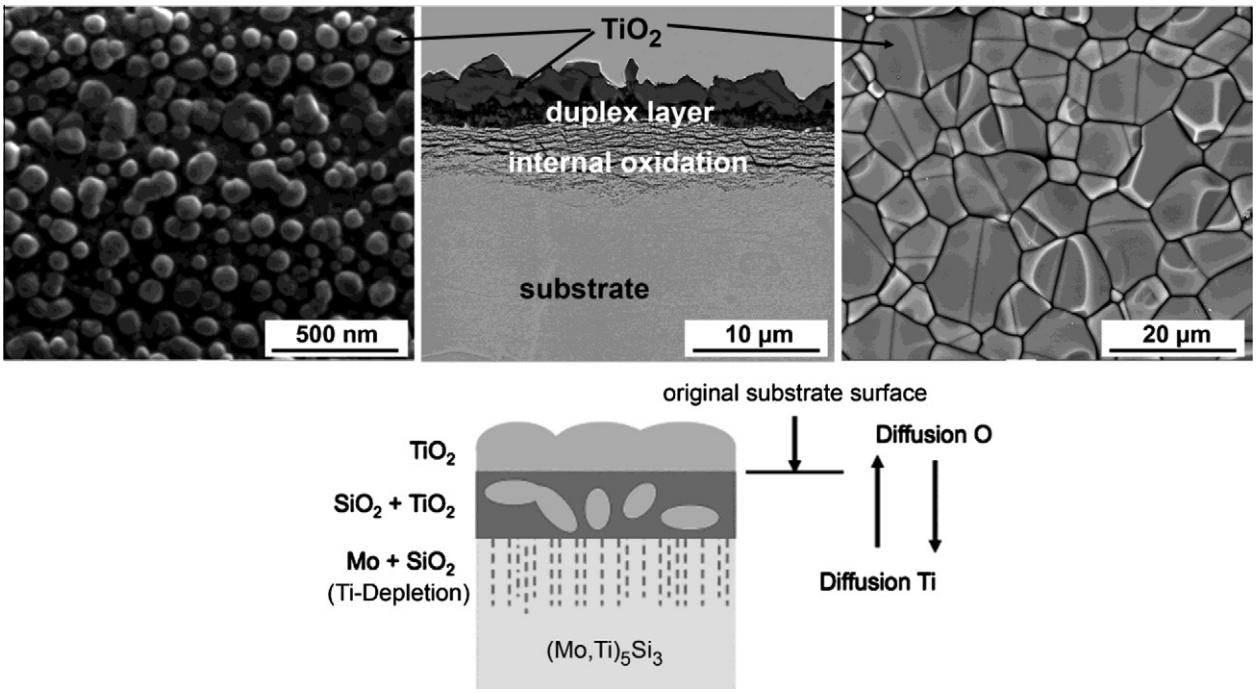


Figure 3. SEM microstructures of $(Mo,Ti)_5Si_3$ oxidized in air for 100 h at 750 °C (top left, surface, SE mode), 1100 °C (top middle, cross-section, BSE mode) and 1300 °C (top right, surface, BSE mode), and proposed oxidation schematic in cross-section view (bottom).

are also summarized schematically in Figure 3 (bottom). It is assumed that external growth of the TiO_2 layer originates from the substrate surface, since Ti diffusion becomes oxidation rate controlling at temperatures beyond 900 °C [9]. The SiO_2–TiO_2 duplex layer underneath is believed to grow mainly by inward diffusion of oxygen as confirmed by Melsheimer et al. using Pt marker experiments [10]. With increasing time and/or temperature, rapid outward diffusion of Ti leads to its depletion within the internal oxidation zone.

The continuous oxidation experiments (see Fig. 2) show good oxidation resistance for the $(Mo,Ti)_5Si_3$ phase in the temperature range of 750–1300 °C through protective oxide scale formation. Extensive EDS analyses at the titania scale/air as well as at the titania/duplex scale interface did not confirm the presence of Mo, and hence it is assumed that volatile MoO_3 formation is not an issue for this alloy during long-term oxidation. The oxygen partial pressure at the duplex-scale/substrate interface is concluded to be below the dissociation pressure of any Mo-oxide formation. However, a combined oxidation behaviour of evaporation of MoO_3 (weight loss) and formation of silica and titania (weight gain) during the initial stage of oxidation cannot be excluded entirely, although the typical condensed Mo-oxide-rich reaction products which are usually located at the reaction chamber walls of the TGA system as a consequence of an evaporation–condensation reaction were not observed.

The advantage of the $(Mo,Ti)_5Si_3$ is most clearly understood by directly comparing its oxidation characteristics with those of single-phase Mo_5Si_3 and Ti_5Si_3. Mo_5Si_3 exhibits no oxidation resistance at temperatures below 1400 °C due to the formation of a highly porous SiO_2 scale which allows oxygen to permeate readily through the scale [6,11,12]. The $(Mo,Ti)_5Si_3$ phase is superior to Mo_5Si_3 with regard to oxidation resistance, and moreover it exhibits a lower density. Furthermore, the $(Mo,Ti)_5Si_3$ phase shows far better oxidation resistance as compared to stoichiometric Ti_5Si_3. Breakaway oxidation, which was observed for this phase at temperatures beyond 1000 °C [13,14], is not observed if Mo is present in the alloy. For stoichiometric Ti_5Si_3, Williams and Akinc assume almost identical Gibbs free energy values for silica and titania, allowing SiO_2 as well as TiO_2 nuclei to form simultaneously on the surface in the initial oxidation stage. With increasing oxidation time, the oxidation behaviour shifts to a breakaway behaviour [7]. The reason is believed to lie in atmospheric nitrogen allowing the formation of a metastable TiN transient layer at the substrate interface, facilitating

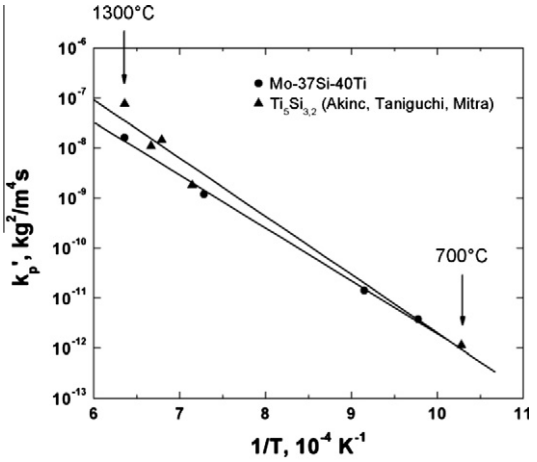


Figure 4. Calculated values of the parabolic oxidation rate constant k_p' of the alloy Mo–37Si–40Ti at temperatures of 700–1300 °C compared to the available literature data for $Ti_5Si_{3.2}$ [10–12].

continuous formation of TiO_2 [14]. The formation of TiN may additionally be enhanced by high diffusivities of nitrogen in TiO_2 [9]. A durable improvement of the oxidation behaviour of Ti_5Si_3 was achieved by adding small amounts of C to the alloy system. It is thought that C reduces oxygen diffusion in the interstitial lattice.

The same effect was achieved by shifting the chemical composition of the alloy to an excess of Si ($Ti_5Si_{3.2}$). Both strategies result in the formation of a SiO_2–TiO_2 duplex layer with SiO_2 as matrix phase, which is also the case for the $(Mo,Ti)_5Si_3$ phase as presented in this work. If the duplex-layer exhibits a TiO_2 matrix, which is the case for the conventional Ti_5Si_3 phase, breakaway oxidation is observed [7].

Figure 4 shows the calculated values of the weight change based parabolic oxidation rate constant k_p' for the alloy composition Mo–37Si–40Ti compared to the available literature data of the $Ti_5Si_{3.2}$ phase, which also oxidizes according to a parabolic relationship. The calculated values of k_p' are similar to the literature values, although k_p' for the $(Mo,Ti)_5Si_3$ phase at 1300 °C is almost one order of magnitude lower than the reference values, underlining its superior oxidation resistance especially at ultra-high temperatures.

The aim of the present investigation was to gain an initial understanding of the oxidation behaviour of the bulk monolithic $(Mo,Ti)_5Si_3$ phase. The main conclusions can be summarized as follows. Study of the alloy with composition Mo–37Si–40Ti shows that beneath a superficial TiO_2 scale a duplex layer containing a mixture of SiO_2 and TiO_2 is formed. It appears to be very important that SiO_2 forms the matrix of this SiO_2–TiO_2 duplex layer. As a consequence an effective oxidation barrier which reduces the continuous growth of an outer TiO_2 layer and inhibits the formation and evaporation of volatile Mo oxide exists. Hence, the intermetallic phase is superior with regard to oxidation resistance compared to both monolithic Mo_5Si_3 and monolithic Ti_5Si_3. Additional studies on the stability of the oxide scales during cyclic oxidation are necessary to provide further clarification of the applicability of the observed oxide scale formation mechanism to severe service conditions.

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